Analysis 2
Serie 12
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1.

We have:

$$(\gamma \star \sigma)(t) := \begin{cases} \sigma(2t) & \text{for } 0 \leq t \leq \frac{1}{2} \\ \gamma(2t-1) & \text{for } \frac{1}{2} \leq t \leq 1 \end{cases}$$

Using the definition of line integrals

$$\begin{aligned} \int_{\gamma \star \sigma} F \cdot d(\gamma \star \sigma) &= \int_0^1 F(\gamma \star \sigma(t)) \cdot (\gamma \star \sigma)'(t) dt \\ &= \int_0^{\frac{1}{2}} F(\sigma(2t)) 2\sigma(2t)'(t) dt + \int_{\frac{1}{2}}^1 F(\gamma(2t-1)(t)) 2\gamma(2t-1)' dt \\ &= \int_0^1 F(\sigma(u) + \sigma'(u)) du + \int_0^1 F(\gamma(z) + \gamma'(z)) dz \\ &= \int_\sigma F \cdot d\sigma + \int_\gamma F \cdot d\gamma \end{aligned}$$

2.

2.a.

Define a homotopy H

$$H : [0,1] \times [0,1] \to U$$

Define the points x and p

$$H(0,s) = x \quad H(1,s) = p$$

Then the every path can be written using the homotopy

$$\gamma_s(t) = H(t,s)$$

$$\Rightarrow f(s) = \int_{\gamma_s} F \cdot d\gamma_s = \int_0^1 F(H(t,s)) \cdot H_t(t,s) dt$$

Where $H_i = \frac{\partial H}{\partial i}$
We want to show that $f'(s) = 0$. This is the case if f is well-posed.

$$\begin{aligned} f'(s) &= \frac{d}{ds} \int_0^1 F(H(t,s)) \cdot H_t(t,s) dt \\ &= \int_0^1 \frac{\partial}{\partial s} (F(H(t,s)) \cdot H_t(t,s)) dt \\ &= \int_0^1 \sum_{i=1}^n \frac{\partial}{\partial s} \left(F_i(H) \cdot (H_t)_i \right) dt \end{aligned}$$

Product rule

$$\frac{\partial}{\partial s} \left(F_i(H) \cdot (H_i)_t \right) = \frac{\partial}{\partial s} F_i(H)(H_t)_i + F_i(H)(H_i)_{ts}$$

And chain rule

$$\frac{\partial}{\partial s} (F_i(H)) = \sum_{j=1}^n \frac{\partial F_i}{\partial x_j} (H)(H_s)_j$$

Since $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ the order of j, i does not matter
$\Rightarrow$

$$f'(s) = \int_0^1 \left(\sum_{0<j,i\leq n} \frac{\partial F_i}{\partial x_j}(H)(H_t)_i (H_s)_j + \sum_{0<i\leq n} F_i(H)(H_i)_{ts} \right) dt$$

This means

$$\frac{\partial}{\partial s} F(H(t,s)) \cdot H_t(t,s) = \sum_{0<j,i\leq n} \frac{\partial F_i}{\partial x_j}(H)(H_t)_i (H_s)_j = \frac{\partial}{\partial t} F(H(t,s)) \cdot H_s(t,s)$$

$$\Rightarrow f'(s) = [F(H(t,s)) \cdot H_s(t,s)]_0^1$$

$$H(0,s) = p \quad H(1,s) = x$$
$$H_s(0,s) = H_s(1,s) = 0$$

The points are fixed

$$\Rightarrow f'(s) = 0$$

2.b.

$$f(x) = \int_{\gamma_x} F \cdot d\gamma_x$$

$$f(x') = \int_{\gamma_{x'}} F \cdot \gamma_{x'} = \int_{\gamma_x \star \sigma} F \cdot d(\gamma_x \star \sigma) = \int_{\gamma_x} F \cdot d\gamma_x + \int_\sigma F \cdot d\sigma$$

$$\Rightarrow f(x) - f(x') = -\int_\sigma F \cdot d\sigma$$

2.c.

3.

3.a.

It’s connected but not simply connected. The z-axis is removed from the domain.

3.b.

$$\operatorname{rot}(F) = \nabla \times F = \begin{pmatrix} \frac{d}{dx} \\ \frac{d}{dy} \\ \frac{d}{dz} \end{pmatrix} \times F(x,y,z)$$

$$= \begin{pmatrix} \frac{1}{x^2+y^2} 2y - \frac{2y}{x^2+y^2} \\ \frac{2x}{x^2+y^2} - \frac{2x}{x^2+y^2} \\ \frac{-2(x^2+y^2) - 2x(2(yz-x))}{(x^2+y^2)^2} - \frac{2(x^2+y^2) - 2y(2(xz+y))}{(x^2+y^2)^2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\Rightarrow F$ is conservative

$$\int_{\gamma_1} F \cdot d\gamma_1 = \int_0^{2\pi} F(\cos(t), \sin(t), 0) \cdot (-\sin(t), \cos(t, 0))dt$$
$$= \int_0^{2\pi} (2\sin(t), -2\cos(t), 0) \cdot (-\sin(t), \cos(t), 0)dt$$
$$= \int_0^{2\pi} -2dt = -4\pi$$

3.c.

From stokes $\Rightarrow$ for every oriented area S

$$\int_{\partial S} F \cdot d\gamma = \int_S \mathrm{rot}(F) \cdot dS = 0$$

Since we can get an area which border is equal to the two path γ_1, γ_2 and does not intersect the z-axis:

$$0 = \int_{\partial S} F d\gamma = \int_{\gamma_1} F \cdot d\gamma_1 + \int_{\overline{\gamma_2}} F \cdot d\overline{\gamma_2}$$
$$\Rightarrow \int_{\gamma_1} F \cdot d\gamma_1 = \int_{\gamma_2} F \cdot d\gamma_2$$

3.d.

no

4.

4.a. false

4.b. true

4.c. false

4.d. true

5.

$$F(x,y) = \begin{pmatrix} \lambda x e^y \\ (y + 1 + x^2)e^y \end{pmatrix}$$

Put the function in $\mathbb{R}^3$. Now we can simply calculate $\mathrm{rot}(F')$

$$\mathrm{rot}(F') = \Delta \times F' = \begin{pmatrix} \frac{d}{dx} \\ \frac{d}{dy} \\ \frac{d}{dz} \end{pmatrix} \times \begin{pmatrix} \lambda x e^y \\ (y + 1x^2)e^y \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2xe^y - \lambda x e^y \end{pmatrix}$$

The function F is conservative if $\mathrm{rot}(F) = 0$

$$\Rightarrow 2xe^y = \lambda x e^y \Rightarrow \lambda = 2$$

We want to find a potential φ

$$\Delta\varphi = F$$

$$\varphi(x,y) = \int 2xe^y dx = x^2 e^y + C(y)$$

$$\frac{\partial \varphi}{\partial y} = x^2 e^y + C'(y) = (y + 1 + x^2)e^y$$
$$\Rightarrow C'(y) = (y+1)e^y$$
$$\Rightarrow C(y) = ye^y + c$$
$$\Rightarrow \varphi(x,y) = (y + x^2)e^y + c$$

6.

$$\begin{aligned}
\int_S \operatorname{rot}(F) \cdot dn &= \int_S \begin{pmatrix} -2z \\ 0 \\ -x \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} dS \\
&= \int_S -3xz \cdot dS \\
&= -3 \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin(\varphi)\cos(\theta)\cos(\varphi)\sin(\varphi) d\varphi d\theta \\
&= 0
\end{aligned}$$

$$\int_S \operatorname{rot}(F) n dS = \int_{\partial S} F \cdot d\gamma$$

with

$$\gamma(t) = (\cos(t), \sin(t), 0)$$
$$\gamma'(t) = (-\sin(t), \cos(t), 0)$$
$$F(\gamma(t)) = \sin(t)\cos(t), 0, 1)$$

$$\int_0^{2\pi} (\sin(t)\cos(t), 0, 1) \cdot (-\sin(t), \cos(t), 0) dt = \int_0^{2\pi} -\sin^2(t)\cos(t) dt$$

cos and sin are 2π periodically

$$\int_S \operatorname{rot}(F) \cdot dn = 0$$

7.

7.a.

$$\operatorname{rot}(\varphi f) = \nabla \times (\varphi f) = (\nabla \varphi) \times f + \varphi \nabla f = (\nabla \varphi) \times f + \varphi \operatorname{rot}(f)$$

7.b.

$$\operatorname{div}(\varphi f) = \sum_{i=1}^{3} \frac{\partial}{\partial x_i}(\varphi_i f_i) = \sum_{i=1}^{3} \frac{\partial \varphi}{\partial x_i} f_i + \varphi \frac{\partial f_i}{\partial x_i} = \langle \nabla \varphi\ ,\ f \rangle + \varphi \operatorname{div}(f)$$

7.c.

$$\operatorname{div}(\operatorname{rot}(f)) = \partial_x\left(\partial_y f_z - \partial_z f_y\right) + \partial_y(\partial_z f_x - \partial_x f_z) + \partial_z\left(\partial_x f_y - \partial_y f_x\right) = 0$$

7.d.

$$\operatorname{rot}(\nabla\varphi) = \begin{pmatrix} \partial_y\partial_z\varphi - \partial_z\partial_y\varphi \\ \partial_z\partial_x\varphi - \partial_x\partial_z\varphi \\ \partial_x\partial_y\varphi - \partial_y\partial_x\varphi \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

8.

$(a) = -(b)$

9.

9.a.

let

$$\omega = d\psi^1 \wedge \dots \wedge d\psi^n$$

Each $d\psi^i$ is a function of $(x_1, \dots, x_n)$

$$d\psi^j = \sum_{i=1}^{n} \frac{\partial\psi^j}{\partial x_i} dx^i$$

$$\omega = \sum_{i_1=1}^{n} \frac{\partial\psi^1}{\partial x_{i_1}} dx^{i_1} \wedge \dots \wedge \sum_{i_n=1}^{n} \frac{\partial\psi^n}{\partial x_{i_n}} dx^{i_n}$$

Since $dx^i \wedge dx^i = 0$ only permutations using all $dx, dx^2, \dots dx^n$ survive.
Denote σ as such a permutation.

$$\sum_{\sigma \in S_n} \left(\prod_{i=1}^{n} \frac{\partial\psi^i}{\partial x_{\sigma(i)}} \right) dx^{\sigma(1)} \wedge \dots \wedge dx^{\sigma(n)}$$

The order matters

$$\begin{aligned} \omega &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \left(\prod_{i=1}^{n} \frac{\partial\psi^i}{\partial x_{\sigma(i)}} \right) dx^1 \wedge \dots \wedge dx^n \\ &= \det\left(\frac{\partial\psi_j}{\partial x_i} \right) dx^1 \wedge \dots \wedge dx^n \end{aligned}$$

9.b.

$$\begin{aligned} \Phi : U \subset \mathbb{R}^n &\to \mathbb{R}^n \\ x &\mapsto \Phi(x) = y \end{aligned}$$

$$\begin{aligned} \int_{\Phi(U)} f(y) dy &= \int_U f(\Phi(x)) |\det(D\Phi(x))| dx \\ &= \int (U) f(\Phi(x)) \left| \det\left(\frac{\partial\varphi_j}{\partial x_i} \right) \right| dx \end{aligned}$$